

Instructor: Date:

Learner: Regional Branch:

Program: Center:

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Let's Start!

Important Elements in Software Testing



Learning Outcome:

Implement and document the software tests, applying trial and error, based on the designed plan and procedures established by the company.

Applies the different types of software testing to the IT solution.

Can Do Statements:

- I can talk about the elements that should be included when testing a software.
- I can describe good test estimation practices.

Ask your partner:

- What would you do if you run out of time to hand in an important project?
- Is it important to have a B plan when planning a working schedule for a project? Why?





Let's Explore!

Professor Henry Cullen is teaching some principles of software testing.

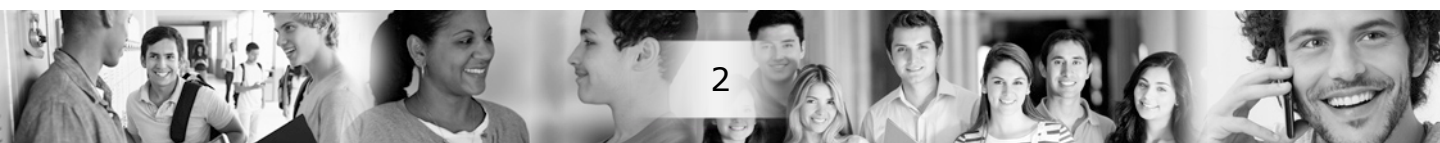


Instructions: Read the text below.

Professor: OK, class. Let's move to page 184. Today, we'll talk about the elements that we need to include when testing software and the importance of each. The core of estimating time is the time frame per se. It's utterly important to stick to the due dates for every stage without putting aside the unpredictable factors. Before starting software testing, break down the project so it will be easier to assign roles. Now, there are four main elements that project leaders need to take into account for planning time in software testing.

The first element is about the resources as they can be physical, human, facilities, and equipment. It's essential to carefully pick out human resources and equipment as they will be the pillars of the project. The second element is time. Allow buffer time, and try to take into account unexpected events that may occur and delay the development of the project. The third element is human skills. A critical resource for any project, so be sure to look for the best and most qualified people since they will help you to achieve your goals and therefore saving you time and money. The last element is cost where you will have to plan the budget for the project.

Now, some recommendations for time estimation is to foresee the bug cycle; this means to bear in mind the errors and the time it takes to fix them. Also, have everything ready; the sufficient resources to keep everything on track. If you have worked before in the previous versions of the software, it will help you estimate the time that every stage requires.





Let's Practice!

Good Estimation Practices

Instructions: Classify the sentences by writing the number of the sentence in the correct category.

1. We'll talk about the elements that we need to include when testing software.
2. There are four main elements that project leaders need to take into account for planning time in software testing.
3. The first element which we describe are the physical, human, facilities, and equipment resources.
4. It's utterly important to stick to the due dates for every stage without putting aside the unpredictable factors.
5. It's essential to carefully pick out human resources and equipment.
6. Before starting software testing, break down the project so it will be easier to assign roles.

Sentences With Relative Clauses	Sentences With Phrasal Verbs

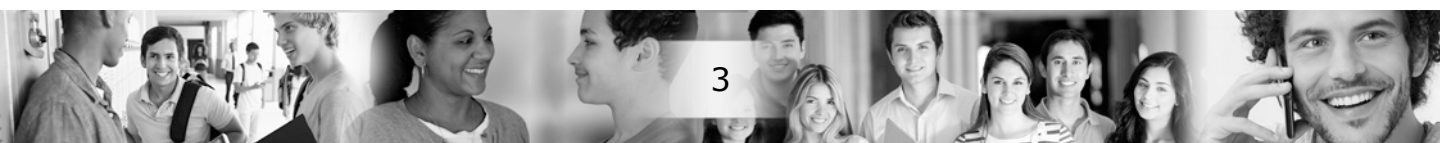


Let's Practice!

Let's test the four software elements.

Instructions: Select the correct answer.

1. There are four main elements
 - a. that project leaders need to take into account.
 - b. that projects leaders need to ignore.
 - c. that projects leaders don't need to remember.
 - d. that project leaders need to teach me.
2. One element is the resources
 - a. which can be mental and emotional.
 - b. which can be sportive and practical.
 - c. which can be physical, human, facilities, and equipment.
 - d. which can be historical and necessary.
3. Another element refers to human skills
 - a. that is very unnecessary for a project.
 - b. that is the only element in a project.
 - c. that is not useful for a project.
 - d. that is a critical resource for any project.
4. The next element is time
 - a. which allows to bear in mind the errors and the time it takes to fix them.
 - b. which allows to control the lunch.
 - c. which allows to bear in mind the errors and the time for dinner.
 - d. which allows to bear in mind the projects and factory.
5. Lastly, the cost
 - a. that is the budget for the project is now for the vacations.
 - b. that is the budget for the project has been increasing every month.
 - c. that is the budget is used for food.
 - d. that is the budget for the project is an unnecessary element.





Extension Activity

Let's talk about estimation practices!

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Draw a poster with each testing software element.
- In each poster, write a brief description of their characteristics.
- Use phrasal verbs and relative clauses.
- Share the four posters with your class and explain them.

Self-Assessment

Yes

No

- I apply the four elements to design an estimation. ☐ ☐
- I understand each element of such estimation. ☐ ☐
- I compare my work to check my understanding of the topic. ☐ ☐
- I use phrasal verbs and relative clauses in a correct way. ☐ ☐



Target Words

- **buffer time** (noun): extra time assigned to a project to avoid readjustment of the original schedule.
e.g. Allow buffer time, and try to take into account unexpected events that may occur and delay the development of the project.
- **budget** (noun): money available to be used for a specific purpose.
e.g. The last element is cost where you will have to plan the budget for the project.
- **foresee** (verb): to predict something.
e.g. Now, some recommendations for time estimation is to foresee the bug cycle.
- **time frame** (noun): a period of time.
e.g. The core of estimating time is the time frame per se.
- **stick to** (verb): to follow something strictly.
e.g. It's utterly important to stick to the due dates for every stage without putting aside the unpredictable factors.
- **break down** (verb): to divide something into steps.
e.g. Before starting software testing, break down the project so it will be easier to assign roles.



Learning Points

Phrasal Verbs

Phrasal verbs are expressions formed with a verb and a preposition to create new and different meanings.

Phrasal verbs are commonly used in English. Whether it is in formal or informal conversations, they make up an important part of the language usage. Most common **phrasal verbs** are **verbs** with **one preposition**, but there are some others with **two** or **three prepositions**. They can also be separable or inseparable, meaning that the object of the sentence can go in between the verb and the preposition. Verbs working with different prepositions may have a different meaning from the sole verb and sometimes they tend to be easily confused. However, it's a matter of practicing. Some **phrasal verbs** are:

Pick out: To **pick** something **out** = to choose

Look for: To **look for** = to search something

Break down: To **break** something **down** = to divide into small parts

Examples:

*We need to **look for** the best workers for our project to be finished on time.*

*We will save time if we **break down** each part of the project.*

Examples

- *Before starting software testing, break down the project so it will be easier to assign roles.*
- *It's essential to carefully pick out human resources and equipment as they will be the pillars of the project.*
- *Be sure to look for the best and most qualified people since they will help you to achieve your goals and therefore saving you time and money.*

Elements to be Estimated in Software Testing

There are crucial factors that require extra planning in software testing that, if planned well, they will make life easier.

Such factors need to work fluently with each other as they are linked in a way that involves all stages and parts of the software to be tested.

There are different ways of calculating *how much* or *how many*. You will need to complete the project but time is something you can't spare or squander. Try to cover all the situations that may happen as your team goes through the project, so in the case of any of those extraordinary situations happen, you won't have to readjust the schedule which will be traumatic to the development of other activities.

Those factors include physical resources such as computers; human resources such as programmers or testers; and facilities such as buildings, offices, and equipment.

The second element is time. You will have to carefully plan the schedule. The third element is human skills in which hiring people who have the right qualities for the project becomes crucial. And the last element is the cost, which includes payment, purchasing, rent, etc.

